

Mikroelektronische Schaltungen und Systeme

Lect.4 Differential Amplifiers

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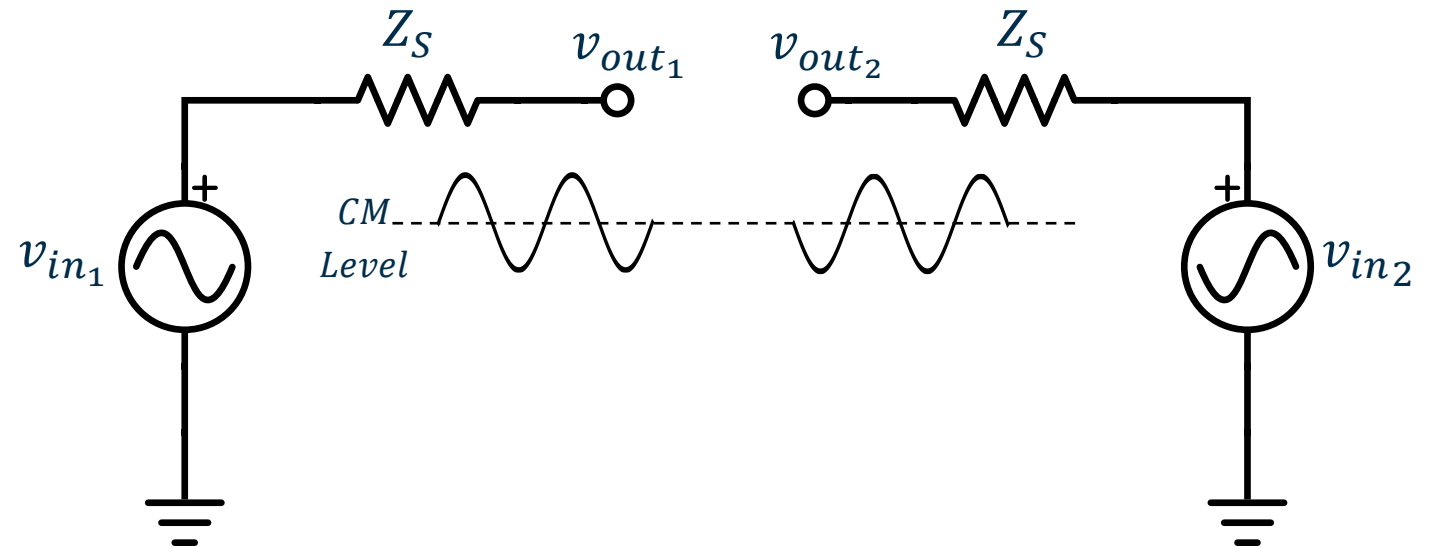
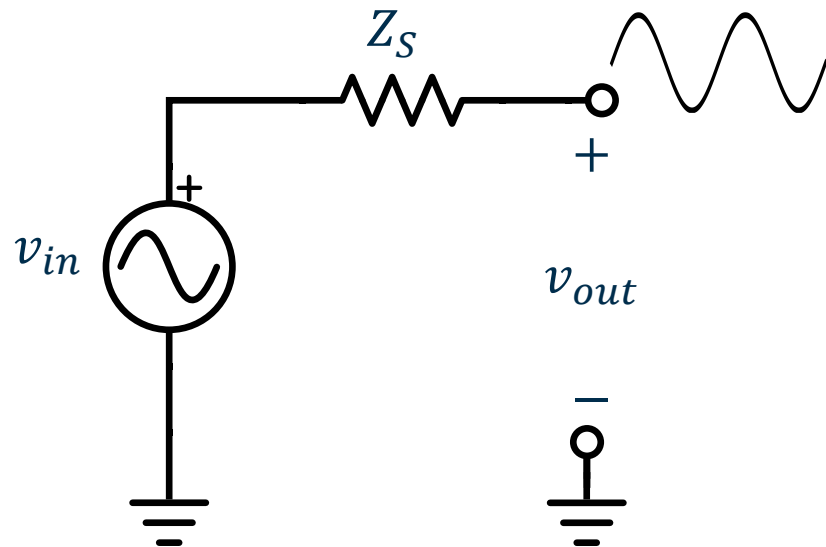
Physical Design II

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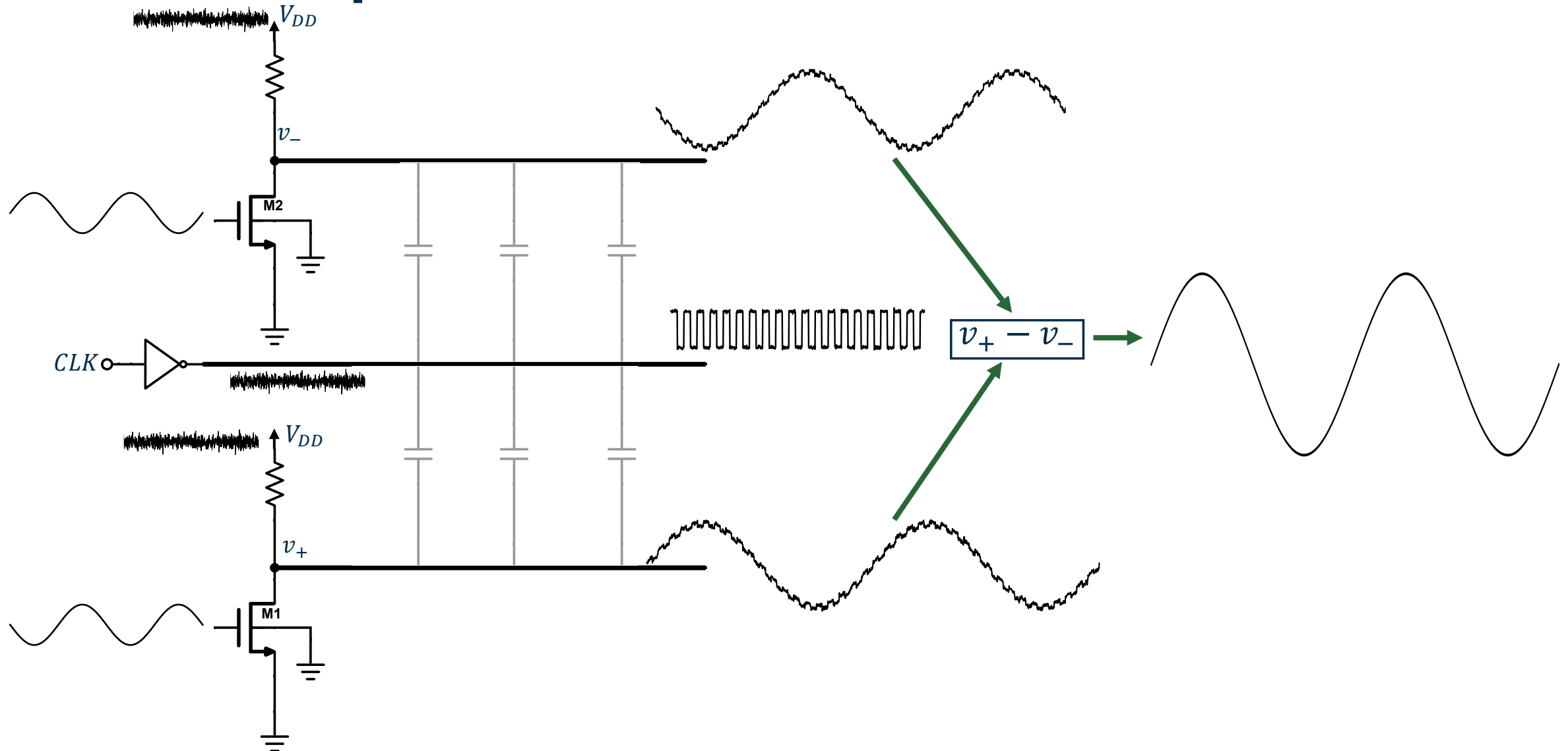
Q&A

Differential Operation

- **Single-Ended Signal:** Measured with respect to a fixed potential, usually ground.
- **Differential Signal:** Measured between two nodes (e.g. v_{out_1} , v_{out_2}) that have equal and opposite excursions around a "common-mode" (CM) level.



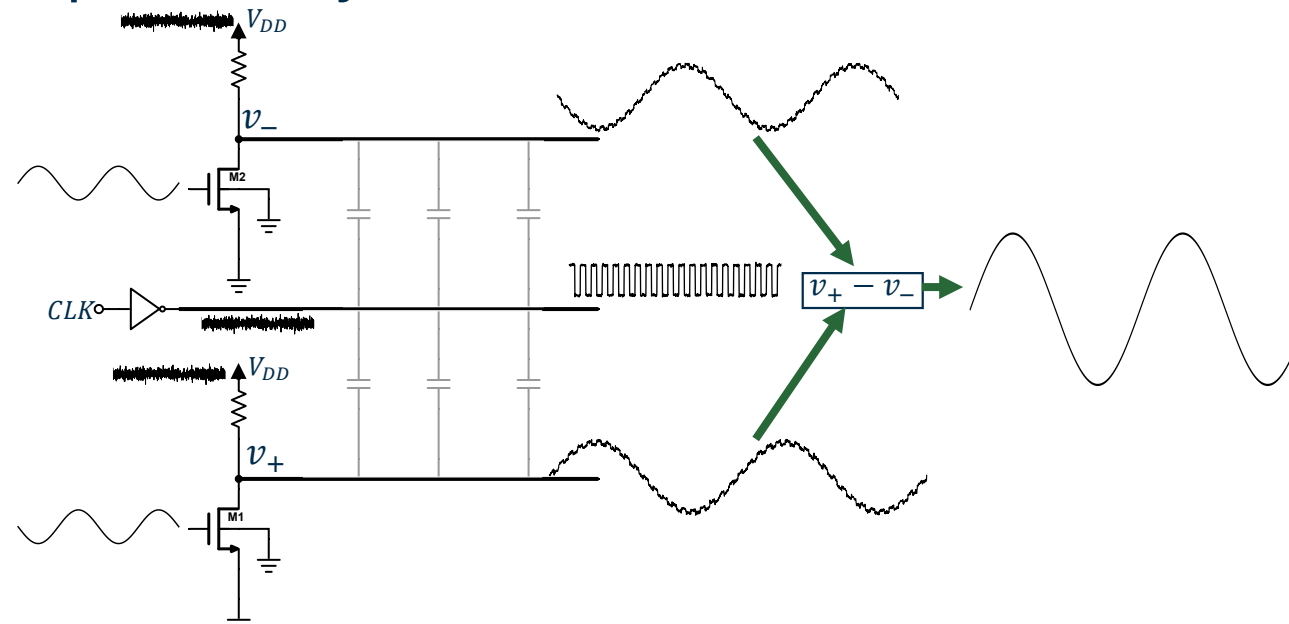
Differential Operation



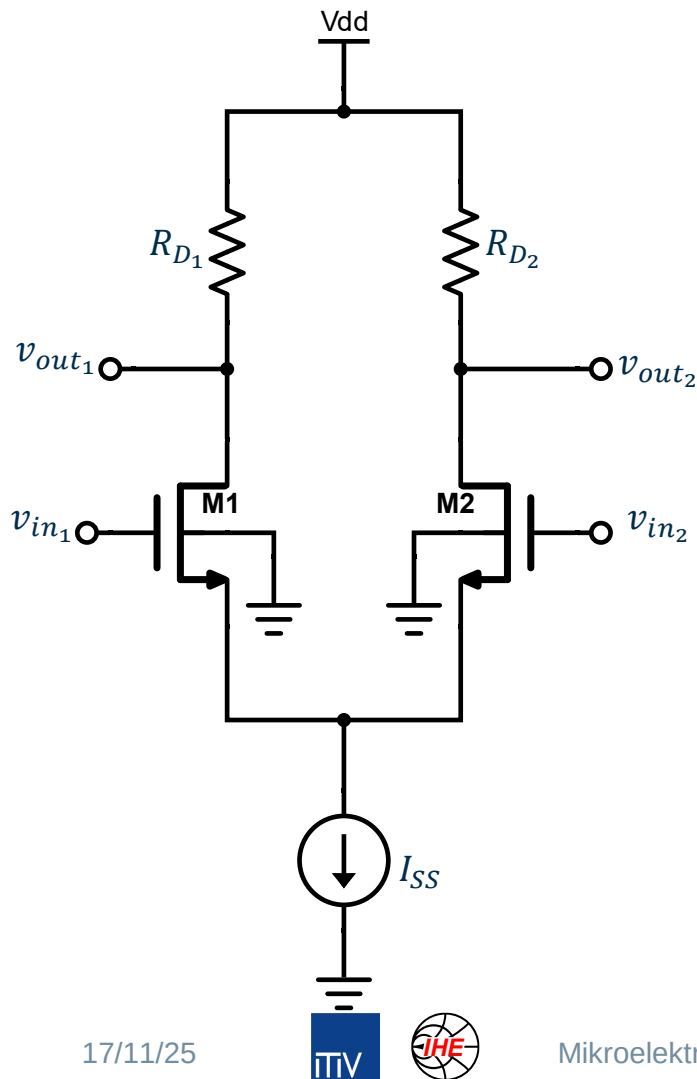
Differential Operation

Why Differential?

- A single-ended signal is highly sensitive to noise or interference (e.g., a clock line) coupling onto it. The amplifier can't distinguish the noise from the signal.
- Differential lines are physically close. The interference couples equally to both lines as a common-mode signal.
- A differential amplifier amplifies only the difference between the lines, rejecting the common-mode noise.

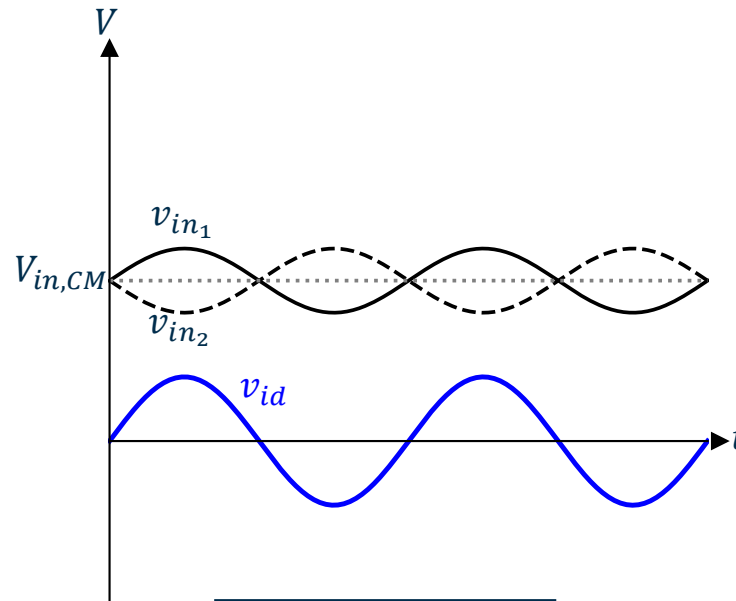
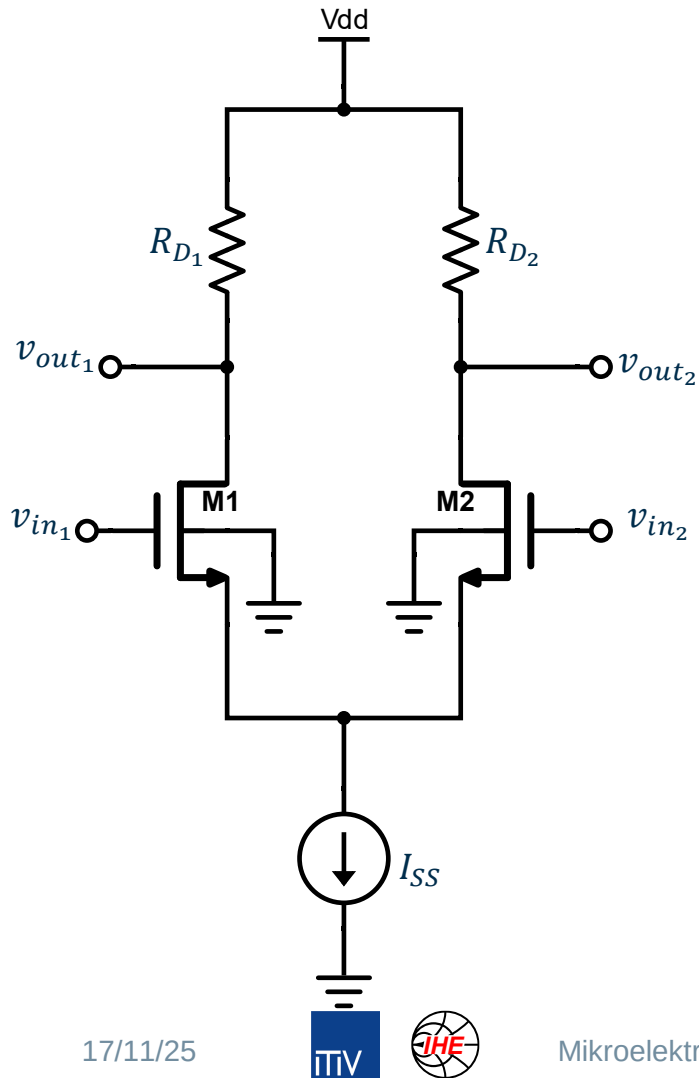


Basic Differential Pair

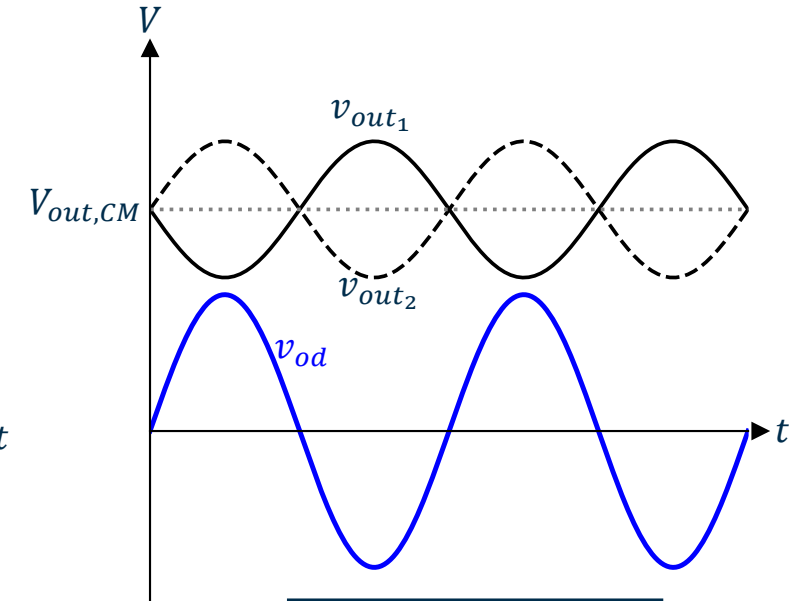


- Two identical single-ended CS amplifiers with differential inputs and a common-mode level are utilized to form a differential pair.
- The output is also differential and has its own common-mode level.
- This topology offers improved noise rejection and larger output swing. However, variations in the input common-mode level can degrade the common-mode rejection ratio (CMRR) and impact overall circuit performance.

Basic Differential Pair

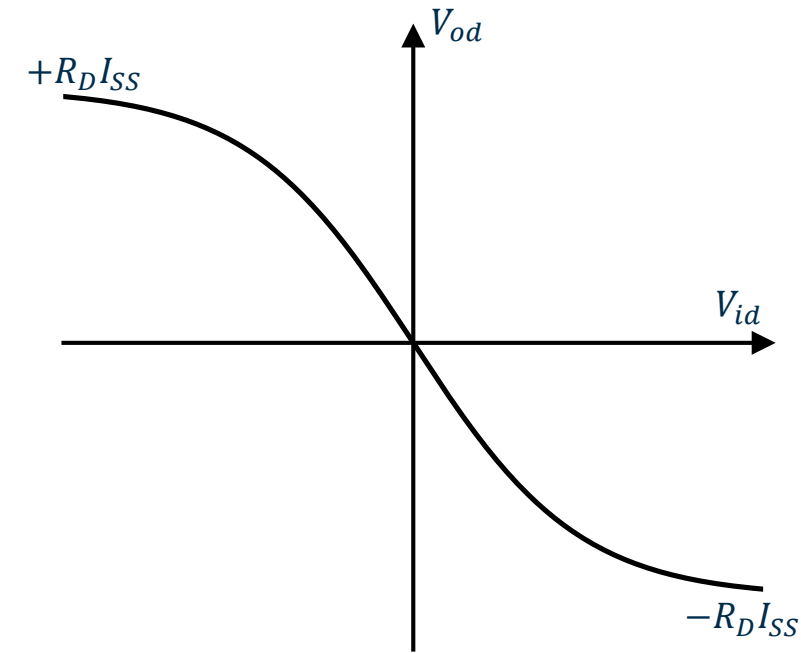
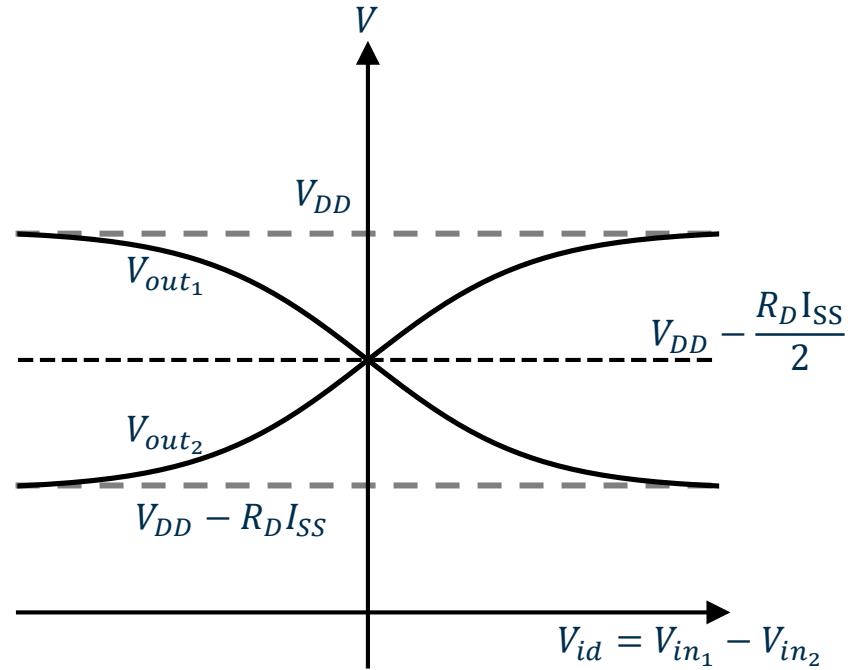
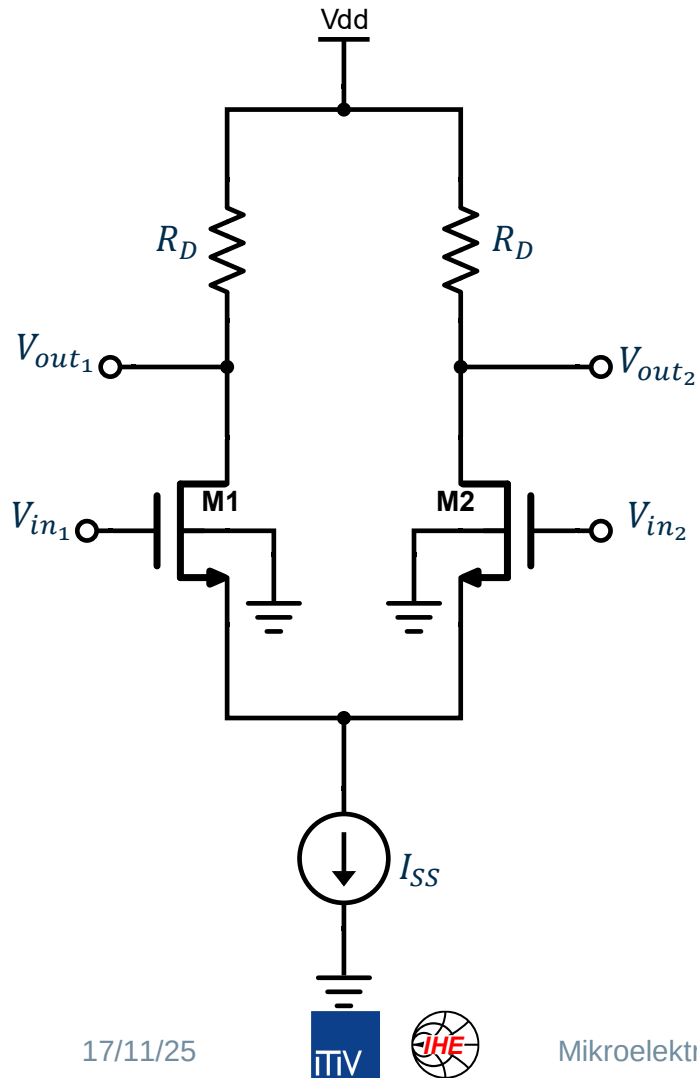


$$\begin{aligned} v_{in_1} &= V_{in,CM} + \frac{v_{id}}{2} \\ v_{in_2} &= V_{in,CM} - \frac{v_{id}}{2} \\ v_{id} &= v_{in_1} - v_{in_2} \end{aligned}$$

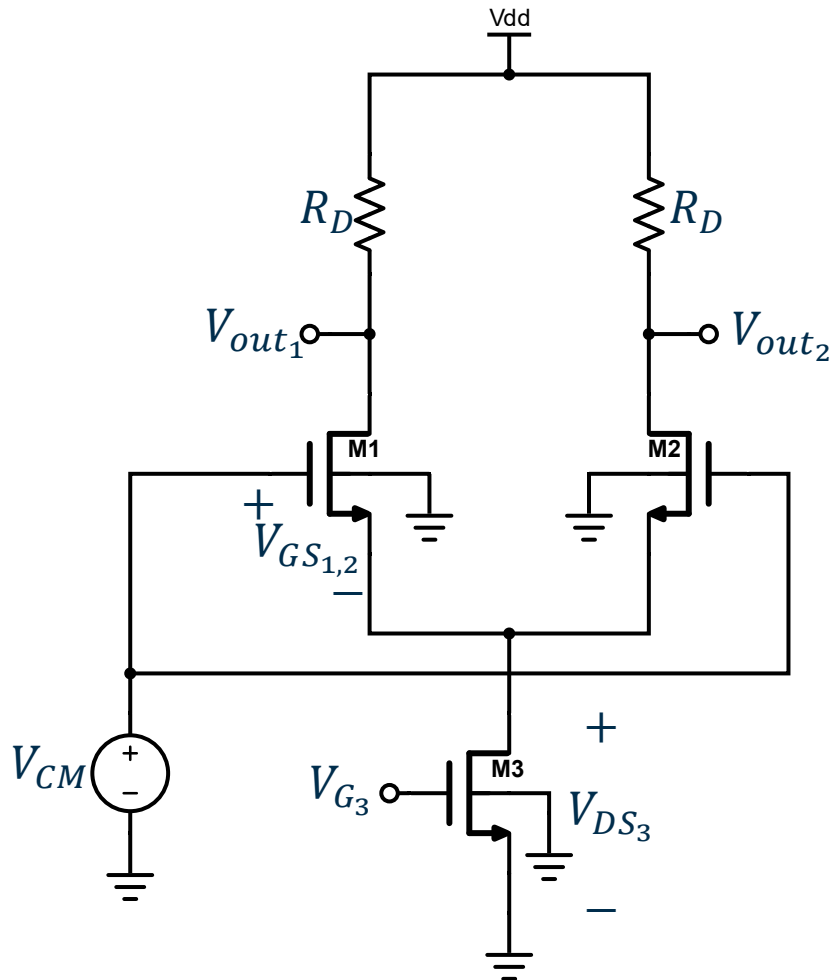


$$\begin{aligned} v_{out_1} &= V_{out,CM} - \frac{v_{od}}{2} \\ v_{out_2} &= V_{out,CM} + \frac{v_{od}}{2} \\ v_{od} &= v_{out_2} - v_{out_1} \end{aligned}$$

Basic Differential Pair



Voltage Swing Boundaries

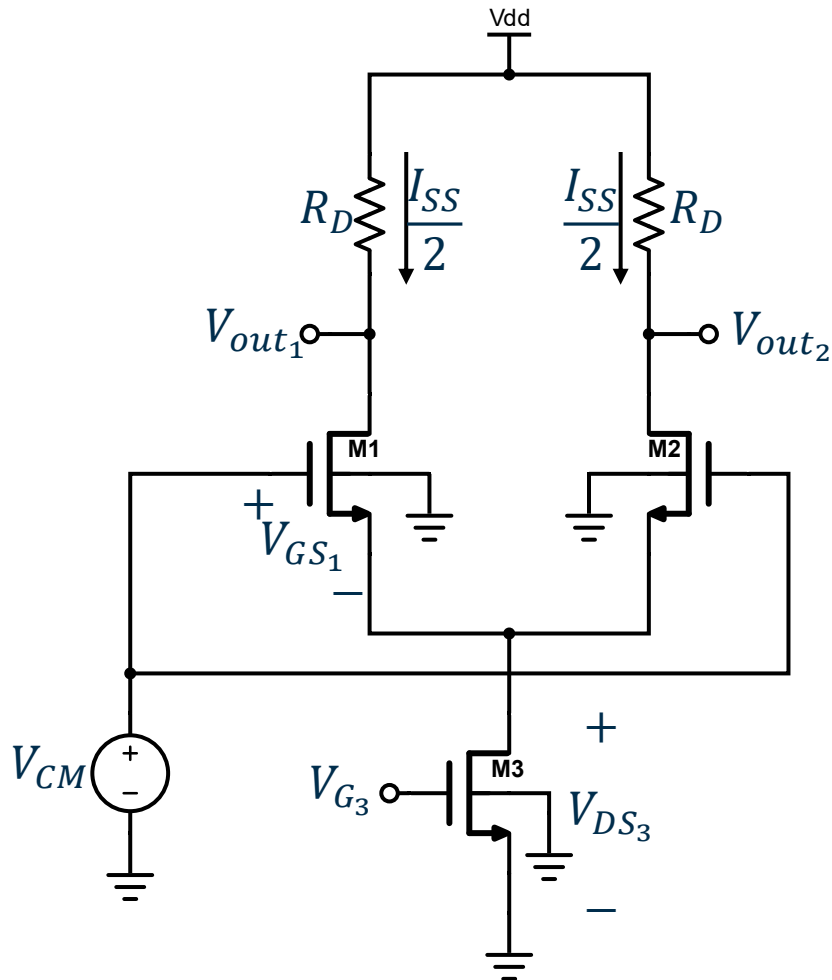


$$V_{DS3} = V_{GS3} - V_{TH3}$$

$$V_{CM} \geq V_{DS3} + V_{GS1}$$

$$V_{CM} \geq V_{GS1,2} + (V_{GS3} - V_{TH3})$$

Voltage Swing Boundaries



$$V_{out1} = V_{DD} - \frac{I_{SS}}{2} R_D, \quad V_{CM} = V_{DS3} + V_{GS1}$$

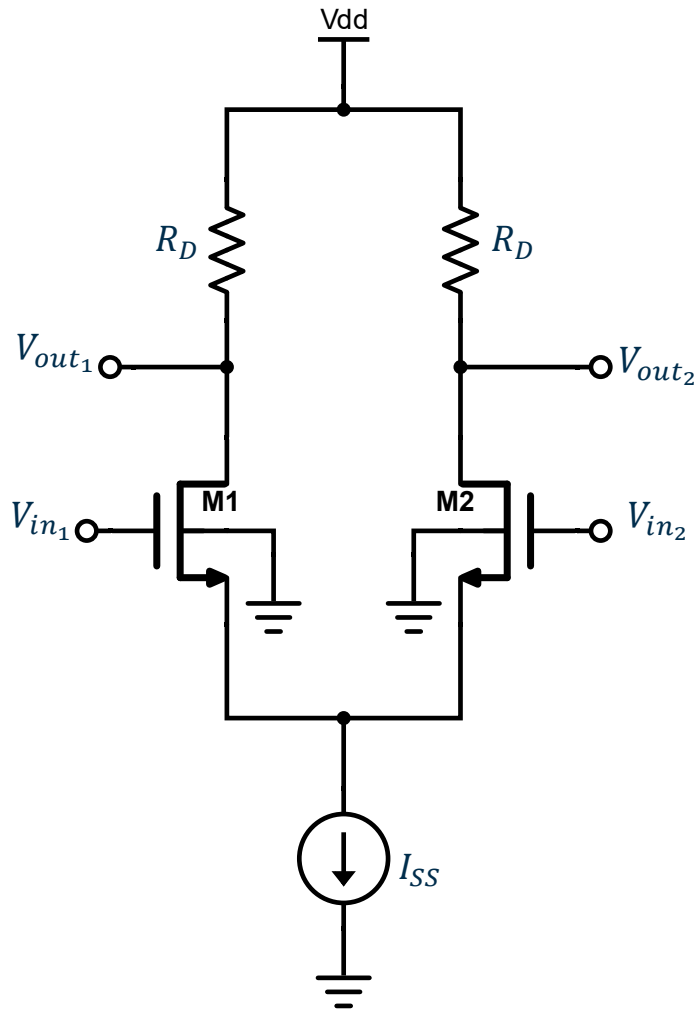
$$V_{out1} - V_{DS3} \geq V_{GS1} - V_{TH1}$$

$$V_{DD} - \frac{I_{SS}}{2} R_D \geq V_{CM} - V_{TH1}$$

$$V_{CM} \leq V_{TH1} + V_{DD} - \frac{I_{SS}}{2} R_D$$

$$V_{GS1} + (V_{GS3} - V_{TH3}) \leq V_{CM} \leq V_{TH1} + V_{DD} - \frac{I_{SS}}{2} R_D$$

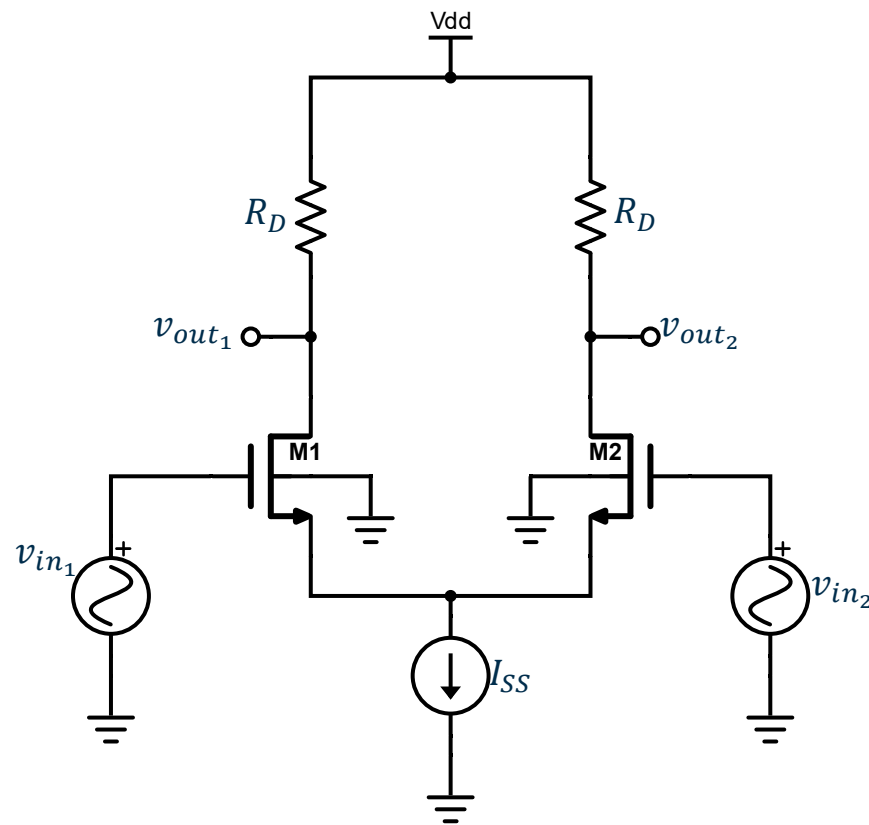
Large Signal Analysis



- $(V_{GS} - V_{TH})^2 = \frac{I_D}{\frac{1}{2}\mu_n C_{ox} \frac{W}{L}} \rightarrow V_{GS} = \sqrt{\frac{2I_D}{\mu_n C_{ox} \frac{W}{L}}} + V_{TH}$
 - $V_{in_1} - V_{in_2} = \sqrt{\frac{2I_{D1}}{\mu_n C_{ox} \frac{W}{L}}} - \sqrt{\frac{2I_{D2}}{\mu_n C_{ox} \frac{W}{L}}}$
 - $(V_{in_1} - V_{in_2})^2 = \frac{2}{\mu_n C_{ox} \frac{W}{L}} (I_{SS} - 2\sqrt{I_{D1} I_{D2}})$
 - $\frac{1}{2}\mu_n C_{ox} \frac{W}{L} (V_{in_1} - V_{in_2})^2 - I_{SS} = -2\sqrt{I_{D1} I_{D2}}$
- $(I_{D1} - I_{D2})^2 = -\frac{1}{4} \left(\mu_n C_{ox} \frac{W}{L} \right)^2 (V_{in_1} - V_{in_2})^4 + I_{SS} \mu_n C_{ox} \frac{W}{L} (V_{in_1} - V_{in_2})^2$
 - $I_{D1} - I_{D2} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{in_1} - V_{in_2}) \sqrt{\frac{4I_{SS}}{\mu_n C_{ox} \frac{W}{L}} - (V_{in_1} - V_{in_2})^2}$
 - $\frac{\partial I_D}{\partial V_{in}} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \frac{\frac{4I_{SS}}{\mu_n C_{ox} \frac{W}{L}} - 2\Delta V_{in}^2}{\sqrt{\frac{4I_{SS}}{\mu_n C_{ox} \frac{W}{L}} - \Delta V_{in}^2}}$
 - For $\Delta V_{in} = 0$, $G_m = \sqrt{\mu_n C_{ox} \frac{W}{L} I_{SS}}$
 - $|A_d| = \sqrt{\mu_n C_{ox} \frac{W}{L} I_{SS} R_D}$

Differential Amplifier Small-Signal Analysis

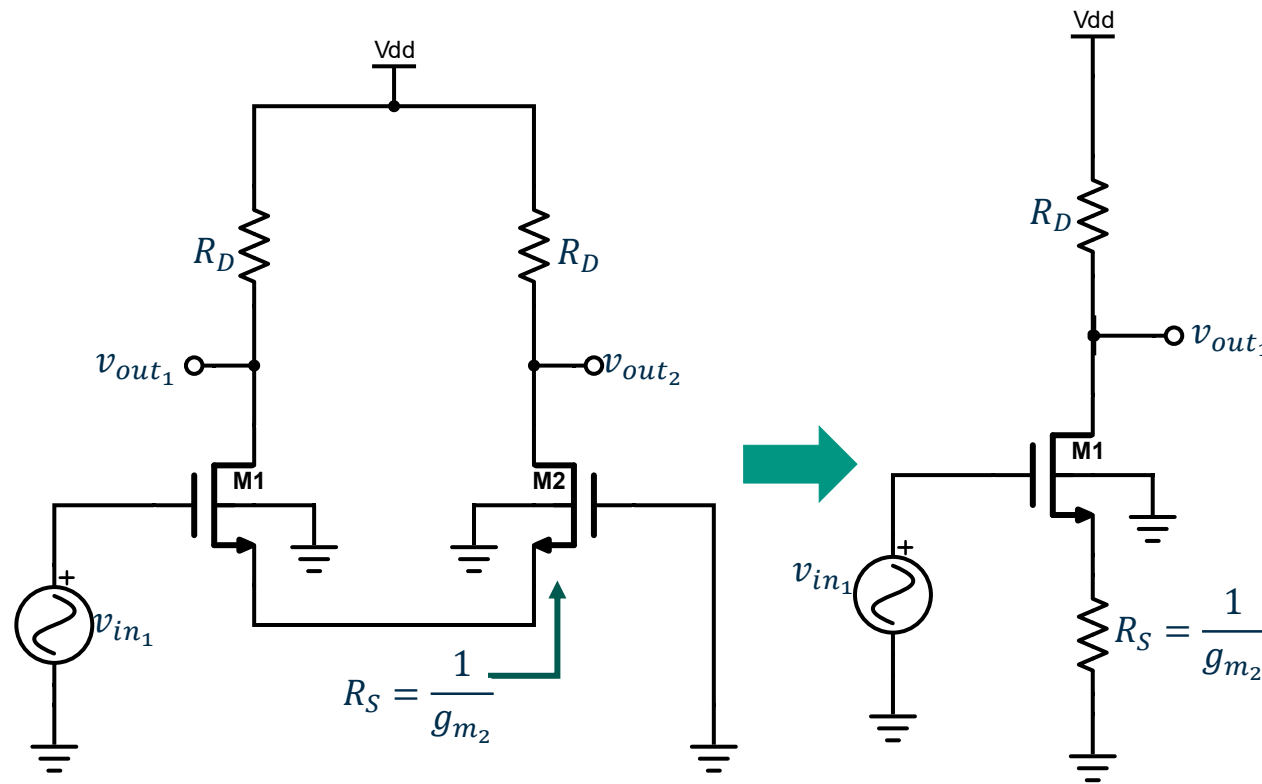
- Small signals v_{in_1} and v_{in_2} are applied as input.
- M1 and M2 are saturated.
- Two methods will be introduced for gain analysis.



Differential Amplifier Small-Signal Analysis

$$g_{m_{eff}} = \frac{g_m}{1 + g_m R_S}$$

Superposition (Only v_{in_1})



$$\frac{v_{out_1}}{v_{in_1}} = -\frac{R_D}{\frac{1}{g_{m_1}} + \frac{1}{g_{m_2}}}, \quad \frac{v_{out_2}}{v_{in_1}} = \frac{R_D}{\frac{1}{g_{m_2}} + \frac{1}{g_{m_1}}}$$

$$(v_{out_1} - v_{out_2}) \Big|_{\text{due to } v_{in_1}} = -\frac{2R_D}{\frac{1}{g_{m_1}} + \frac{1}{g_{m_2}}} v_{in_1}$$

Due to the symmetry, the effect of v_{in_2} at the output ports is identical to that of v_{in_1} except for a change in polarities.

$$(v_{out_1} - v_{out_2}) \Big|_{\text{due to } v_{in_2}} = \frac{2R_D}{\frac{1}{g_{m_1}} + \frac{1}{g_{m_2}}} v_{in_2}$$

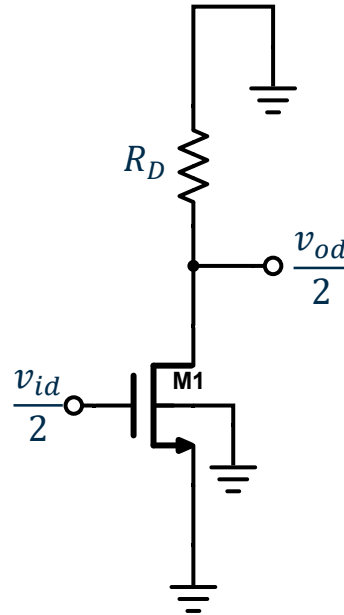
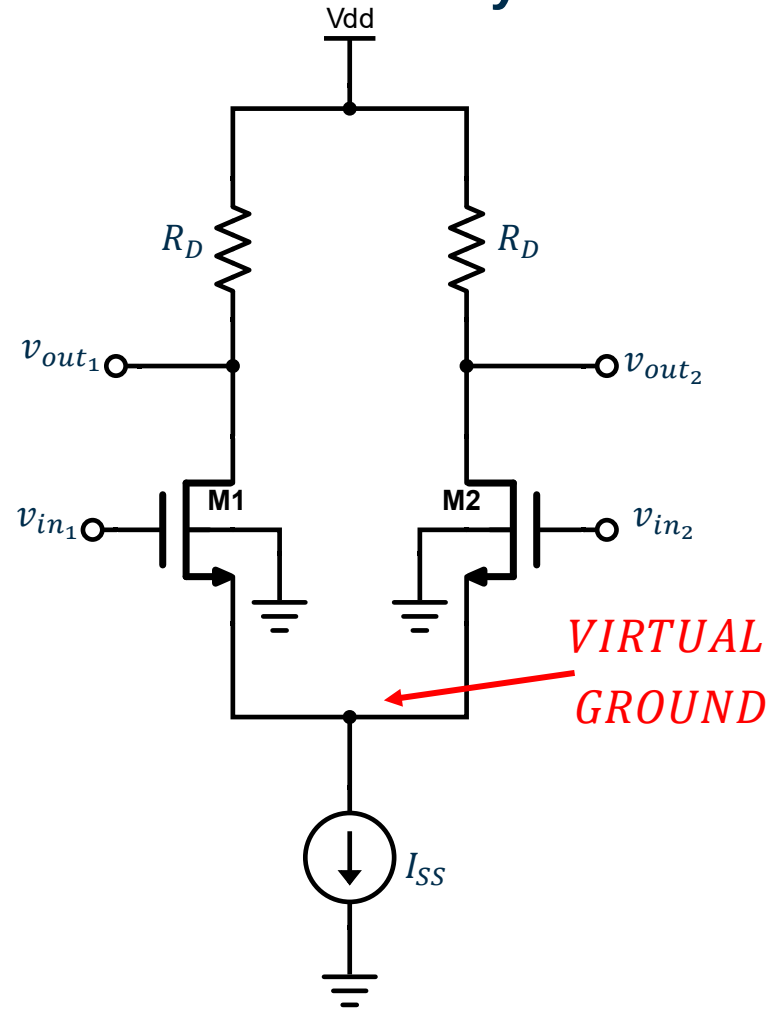
$$\frac{v_{out_1} - v_{out_2}}{v_{in_1} - v_{in_2}} = -\frac{2R_D}{\frac{1}{g_{m_1}} + \frac{1}{g_{m_2}}}$$

For $g_m = g_{m_1} = g_{m_2}$

$$\frac{v_{out_1} - v_{out_2}}{v_{in_1} - v_{in_2}} = -g_m R_D$$

Differential Amplifier Small-Signal Analysis

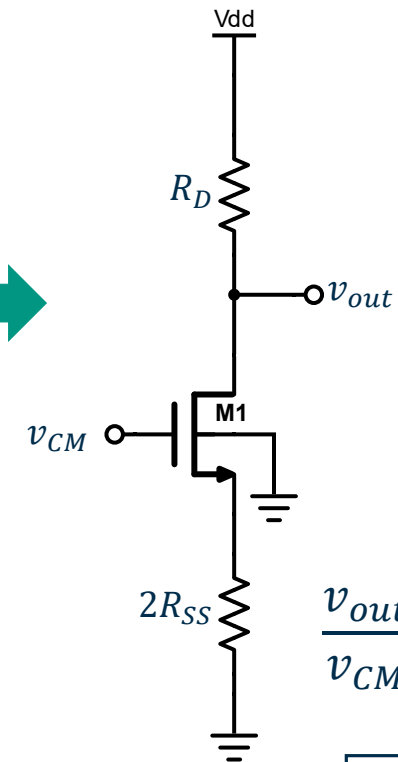
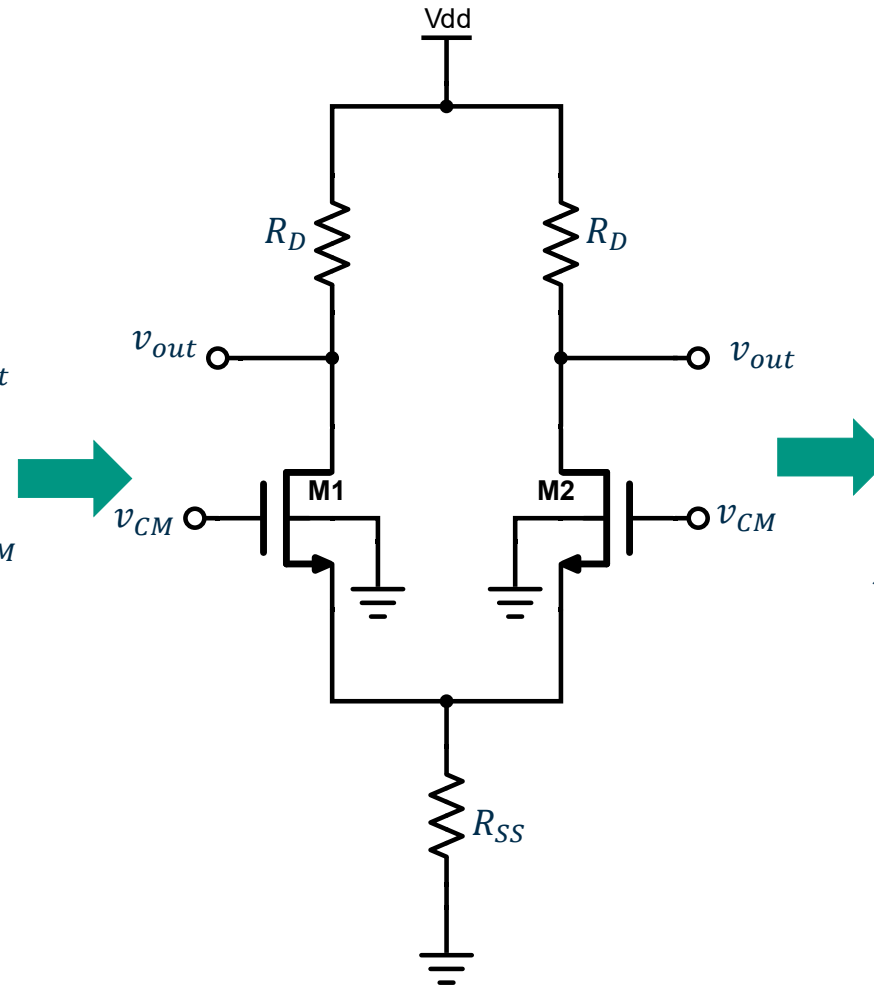
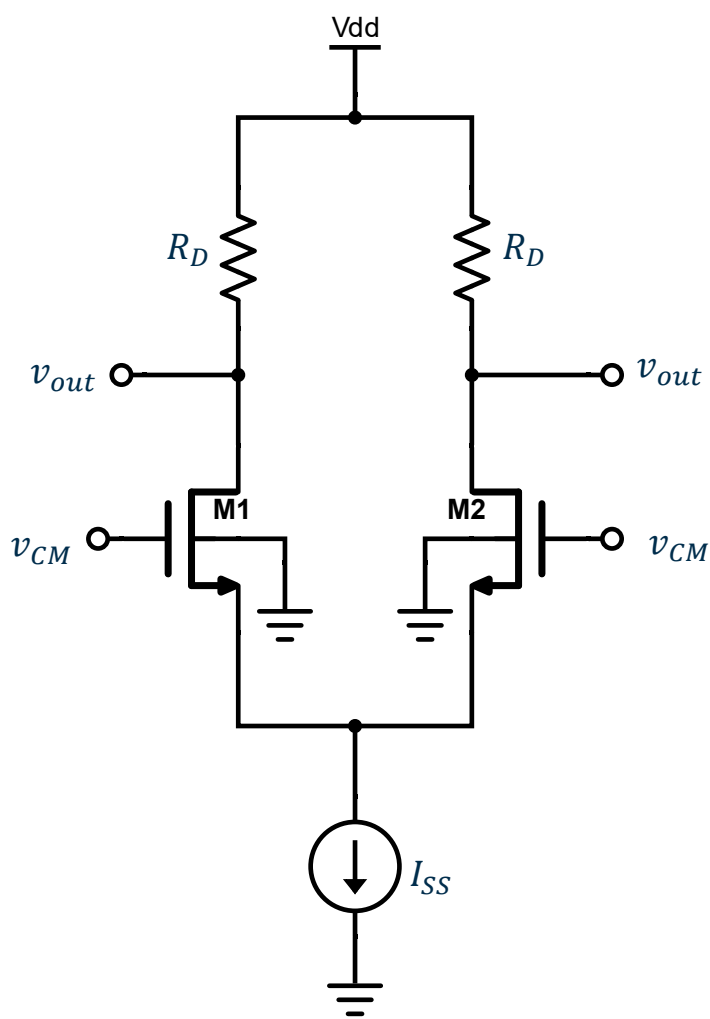
Half Circuit Analysis



$$v_{out1} = -v_{out2} = \frac{v_{od}}{2}$$
$$v_{in1} = -v_{in2} = \frac{v_{id}}{2}$$

$$|A_d| = \left| \frac{v_{od}}{v_{id}} \right| = g_m R_D$$

Common-Mode Response

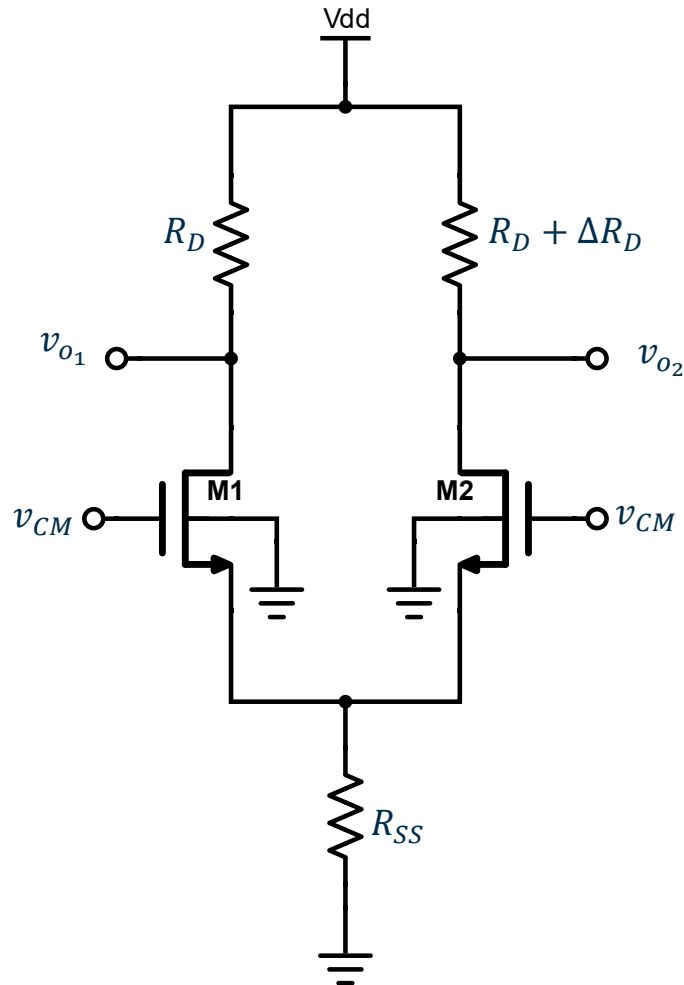


$$\frac{v_{out}}{v_{CM}} = -\frac{g_m}{1 + g_m 2R_{SS}} R_D$$

$$A_{cm} = \frac{v_{out}}{v_{CM}} \approx -\frac{R_D}{2R_{SS}}$$

Common-Mode Response

R_D Mismatch



$$v_{o1} = -\frac{R_D}{2R_{SS}} v_{CM}$$

$$v_{o2} = -\frac{R_D + \Delta R_D}{2R_{SS}} v_{CM}$$

$$v_{od} = v_{o2} - v_{o1} = -\frac{\Delta R_D}{2R_{SS}} v_{CM}$$

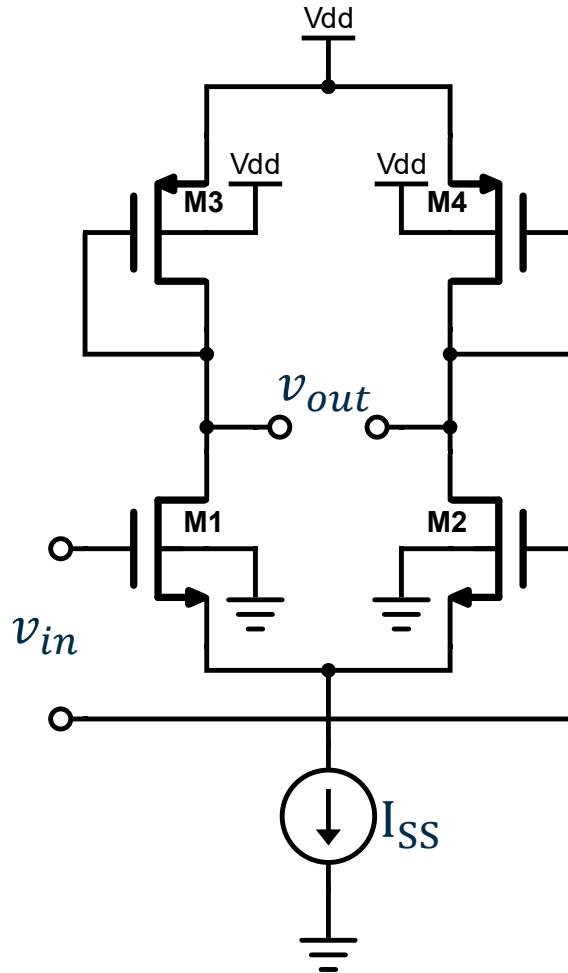
$$A_{cm} = \frac{v_{od}}{v_{CM}} = -\frac{\Delta R_D}{2R_{SS}} = -\left(\frac{R_D}{2R_{SS}}\right)\left(\frac{\Delta R_D}{R_D}\right)$$

Common-Mode Rejection Ratio (CMRR)

$$CMRR = \frac{\text{Differential Gain}}{\text{Common - Mode Gain}} = \frac{|A_d|}{|A_{cm}|} \approx \frac{g_m R_D}{\frac{R_D}{2R_{SS}}} = 2g_m R_{SS}$$
$$CMRR(dB) = 20 \log \left(\frac{|A_d|}{|A_{cm}|} \right)$$

Differential Pair with MOS Loads

Diode-Connected Load



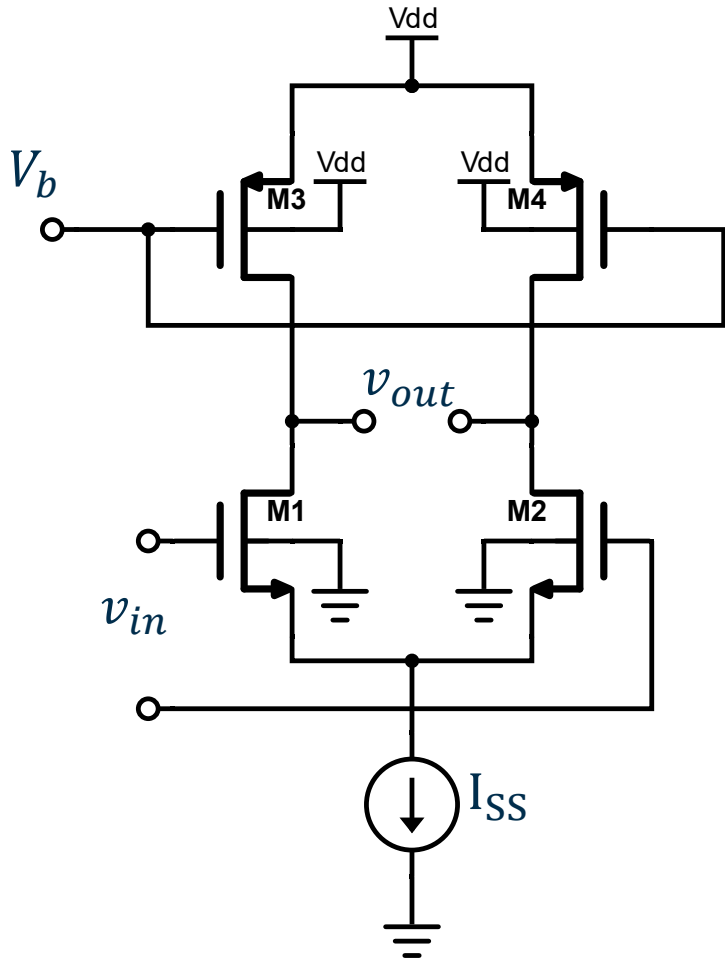
$$A_d = \frac{v_{out}}{v_{in}} = -g_{m_N} \left(\frac{1}{g_{m_P}} \parallel r_{o_N} \parallel r_{o_P} \right)$$

- N and P denote NMOS and PMOS, respectively.

$$A_d = \frac{v_{out}}{v_{in}} \approx -\frac{g_{m_N}}{g_{m_P}} = -\frac{\mu_n \left(\frac{W}{L} \right)_N}{\mu_p \left(\frac{W}{L} \right)_P}$$

Differential Pair with MOS Loads

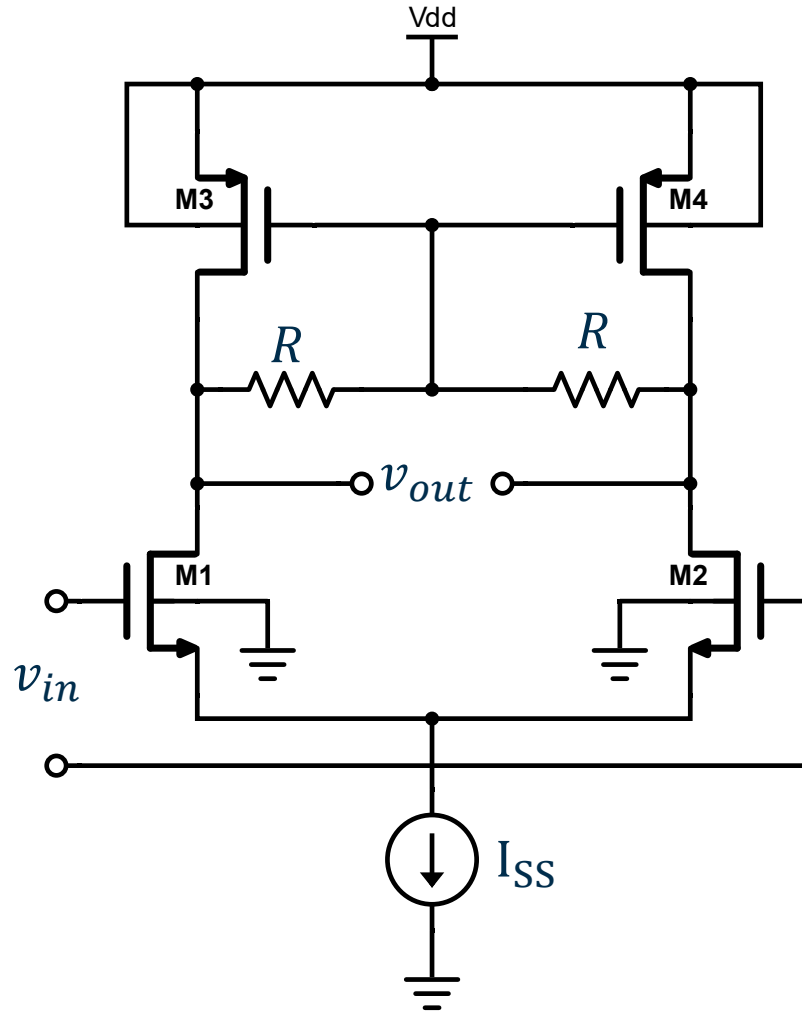
Current-Source Load



$$A_d = \frac{v_{out}}{v_{in}} = -g_{m_N}(r_{o_N} \parallel r_{o_P})$$

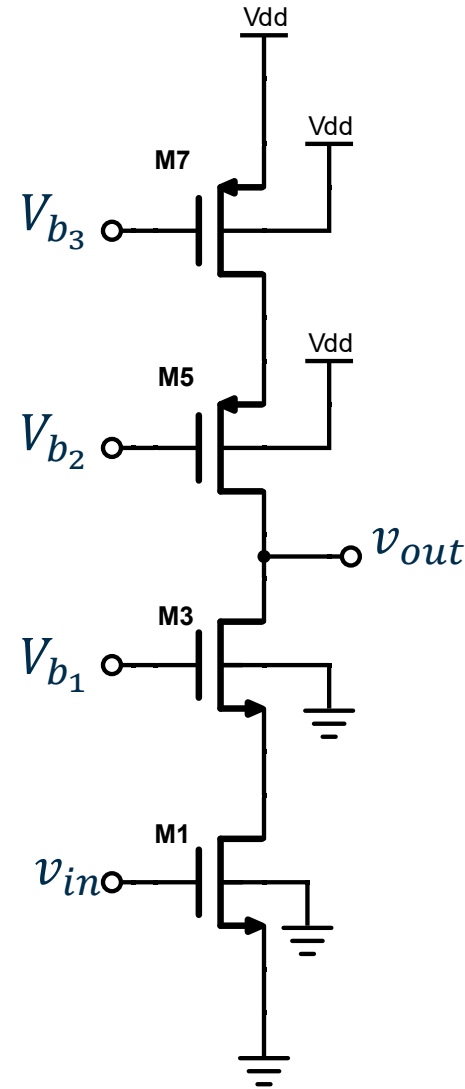
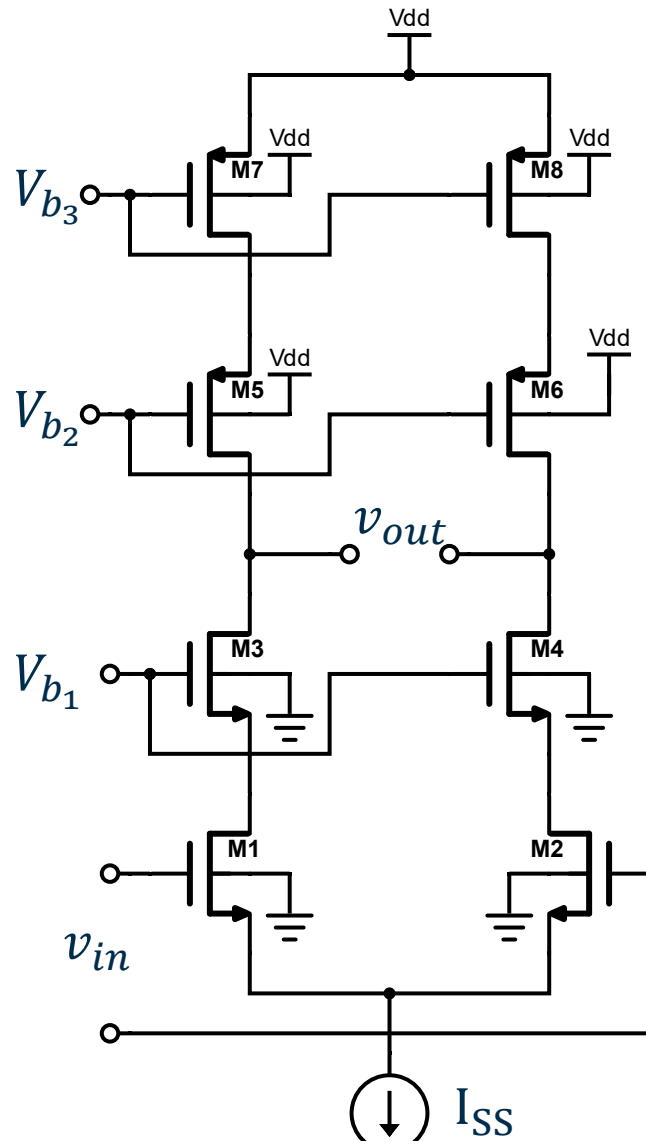
Differential Pair with MOS Loads

Self-Biased Load



$$A_d = \frac{v_{out}}{v_{in}} = -g_{m_1} (r_{o_1} \parallel R \parallel r_{o_3})$$

Cascode Differential Pair



$$|A_d| \approx g_{m_1} [(g_{m_3} r_{o_3} r_{o_1}) \parallel (g_{m_5} r_{o_5} r_{o_7})]$$